

CT-01: Computer Network Time: 40 minutes

1. Class D addresses are designated for multicast. Explain why a typical corporation requiring addresses for its individual workstations and servers cannot use a block of Class D addresses for this purpose. (4 marks)
2. What is the fundamental difference between a *subnet mask* and a *default mask* in IPv4 addressing? Provide an example of each for a Class B network. (4 marks)
3. An ISP has allocated the address block 192.168.40.0/22 to a small business. Which of the following host addresses would be routable within this business's network? Justify your answer. (4 marks)
 - I. 192.168.39.255
 - II. 192.168.41.10
 - III. 192.168.43.1
 - IV. 192.168.40.254
4. A university department is granted the IP block 172.16.32.0/20. They need to create at least 12 subnets, with the largest subnet requiring 150 hosts. (4 marks)
 - I. What is the new subnet mask that allows for this?
 - II. How many usable host addresses will each of these new subnets have?
 - III. What is the network address and broadcast address of the 3rd usable subnet?
5. The TCP/IP model is often compared to the OSI model. In a single, clearly labeled diagram, map the four layers of the TCP/IP model to the seven layers of the OSI model, showing their correspondence. (4 marks)

$$\begin{array}{r} 256 \\ 41 \\ \hline 215 \end{array}$$

$$\begin{array}{r} 32 \\ 17 \\ \hline 25 \\ 28 \\ \hline 256 \end{array}$$

$$\begin{array}{r} 12 \overline{) 256} \\ 24 \\ \hline 16 \end{array}$$

$$\begin{array}{r} 128 \\ 64 \\ 32 \\ \hline 16 \end{array}$$

$$\begin{array}{r} 128-190 \\ 12 \overline{) 256} \\ 24 \\ \hline 16 \end{array}$$

$$\begin{array}{r} 32-64 \\ 21-42 \\ \hline 43-64 \end{array}$$

1 Critically analyze why Message Switching is no longer prevalent in modern communication networks.

Examine its inherent limitations with respect to storage requirements, end-to-end latency, and its unsuitability for real-time communication. Compare these limitations with the advantages offered by Packet Switching to justify the shift in modern network design.

2 Critically compare the Stop-and-Wait and Sliding Window flow control mechanisms.

Analyze their differences in terms of transmission efficiency, bandwidth utilization, and responsiveness to varying network conditions. Evaluate scenarios in which each mechanism would be most appropriate, supporting your answer with practical examples.

3 Analyze Single-bit, Multiple-bit, and Burst errors in digital communication systems.

Compare these error types based on their underlying causes, detection complexity, and impact on data integrity. Illustrate each error type with a realistic example of how it may occur in a communication channel.

4 Examine the role of TCP control flags (SYN, ACK, FIN, RST, PSH, URG) in reliable data communication.

Analyze how these flags contribute to connection establishment, data transfer, connection termination, and error recovery. Illustrate their usage through the TCP three-way handshake and graceful connection termination process.

5. Critically analyze the UDP header structure and the function of each field (Source Port, Destination Port, Length, and Checksum).

Evaluate why the checksum field is optional in IPv4 and discuss the potential reliability and security implications of disabling it in practical networking scenarios.